

APMA 4302: Methods in Computational Science

Spring 2026

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Lecture 01

Clarification of course objectives, intended audience, subject scope and prerequisites

Outline

- Course Objectives
- Syllabus:
 - What this course covers
 - What this course does not cover
- Texts/Resources
- Course logistics
- Course pre-requisites
- Questions

Course Objectives

- Provide a practical introduction to the central concepts, theory, and working environment necessary to effectively use HPC resources to solve scientific problems.
- Provide a high-level survey of current hardware, software, libraries, and software development toolchain for HPC.
- Provide a specific, hands-on experience using PETSc to explore large-scale solutions of numerical PDEs.
- A natural successor to APMA 4300 and 4301
- Required for the CSGF Program of Study.
- Provide one particular roadmap into the very broad field of CSE

Venn Diagram of CSE* and HPC*

*CSE: Computational
Science & Engineering

*HPC: High Performance
Computing

*TLA: Three letter acronym

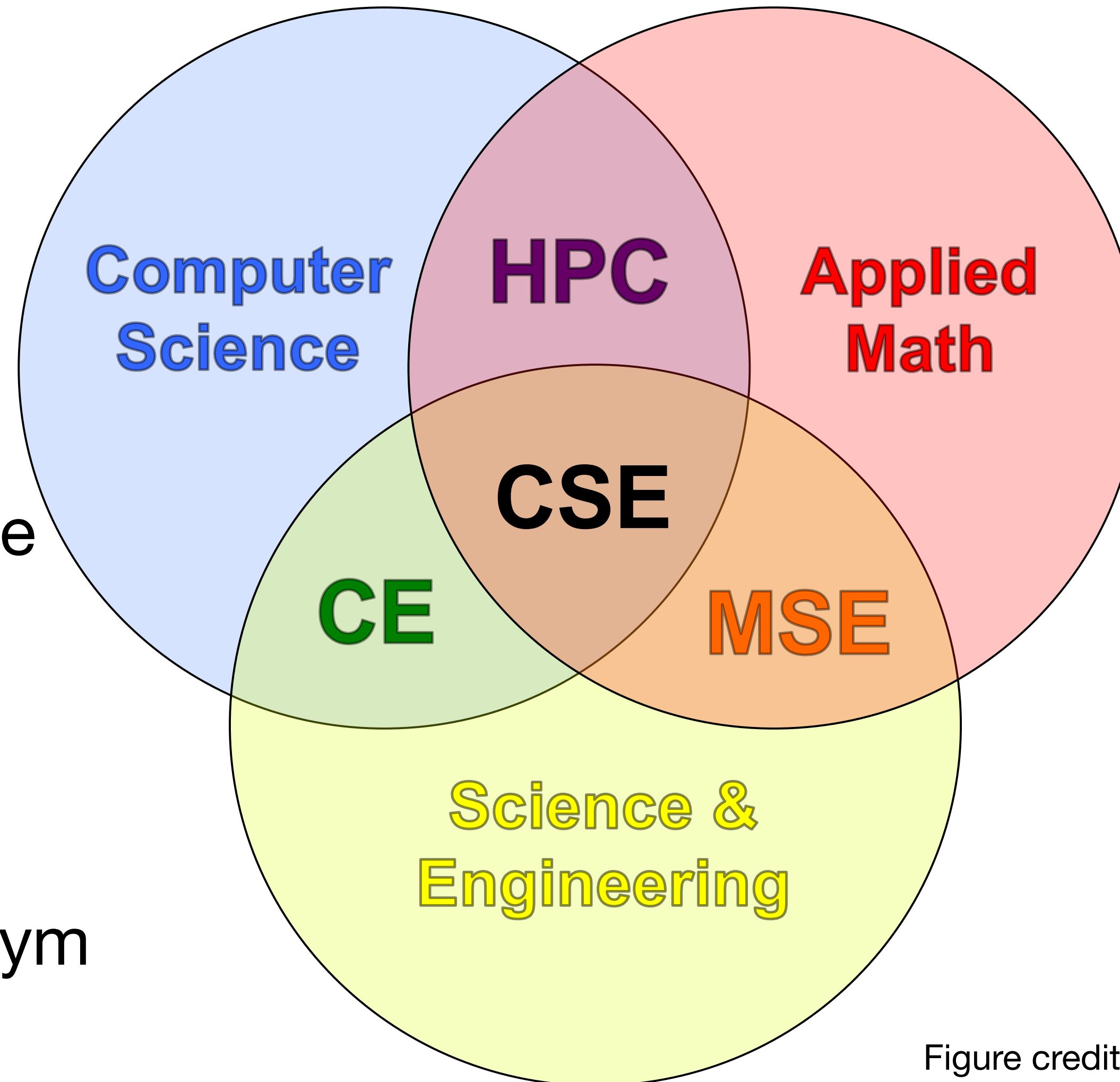


Figure credit: David Keyes

How to choose...



Syllabus

- **Overview of HPC (2-3 lectures)**
 - A brief History of HPC
 - Survey of Hardware
 - Survey of Software
 - The software development/deployment ecosystem and workflows
 - Survey of HPC problems and Kernels
- **Theory and Practice for solving large-scale Numerical PDE's (~20 lectures)**
 - Intro to PETSc/TAO
 - Parallel Linear Algebra
 - Non-linear Algebraic solvers
 - Time Stepping
 - Parallel discretization methods (FD, FV, FEM)
 - Applications to Canonical problems (Poisson, Diffusion, etc)
 - FEM methods with Firedrake
- **Guest Lectures on advanced topics (and/or Project presentations) (~5 lectures)**

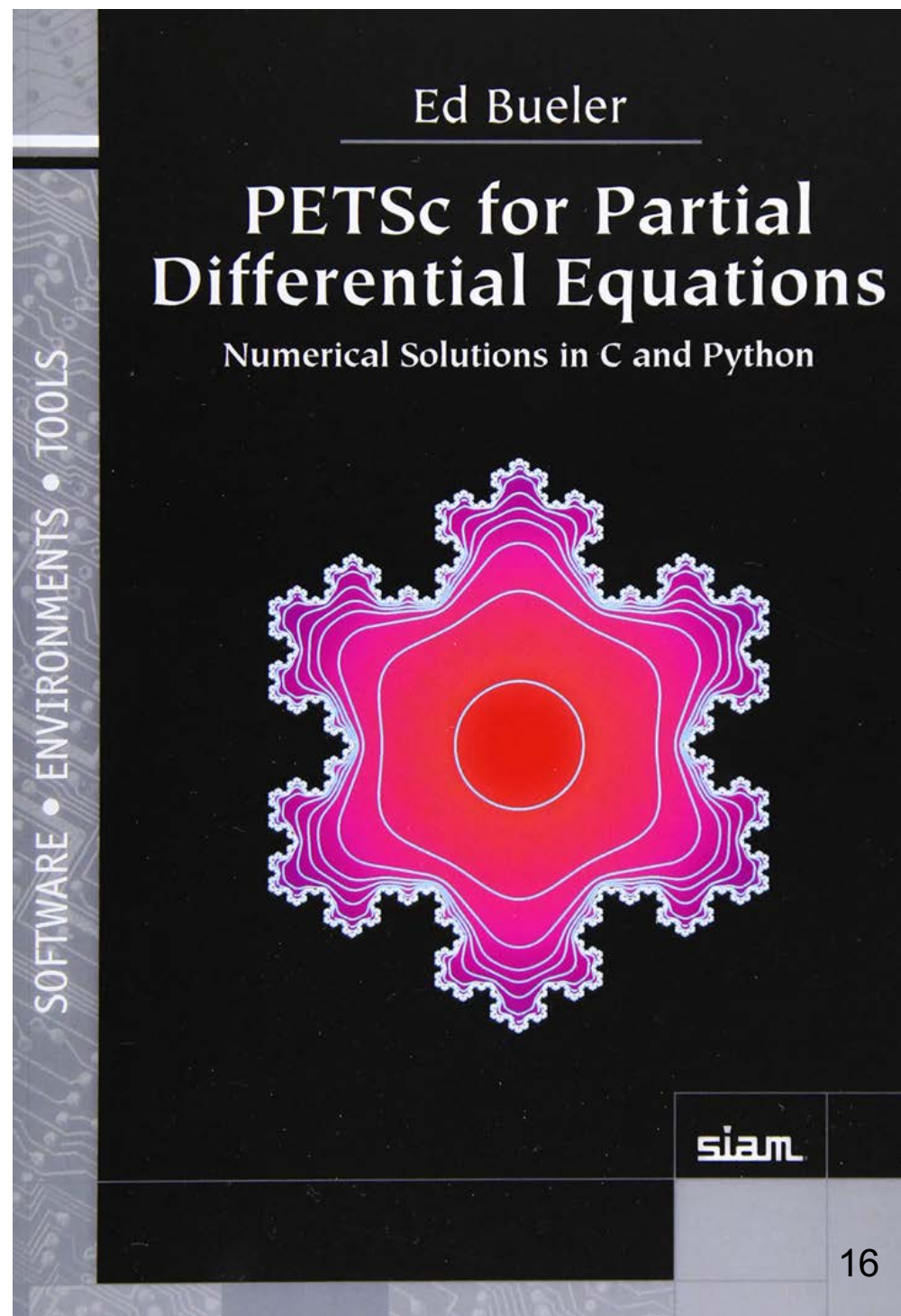
Not covered

- Not a introductory programming class for linux/C/C++
- low level parallel programming per se (MPI, OpenMP, Cuda, HIP, SYCL)
- AI/ML algorithms
- Lots of interesting HPC problems
 - N-Body problems
 - Monte-Carlo methods
 - Molecular Dynamics
 - DFT
 - Something I'm sure you're interested in...

But...

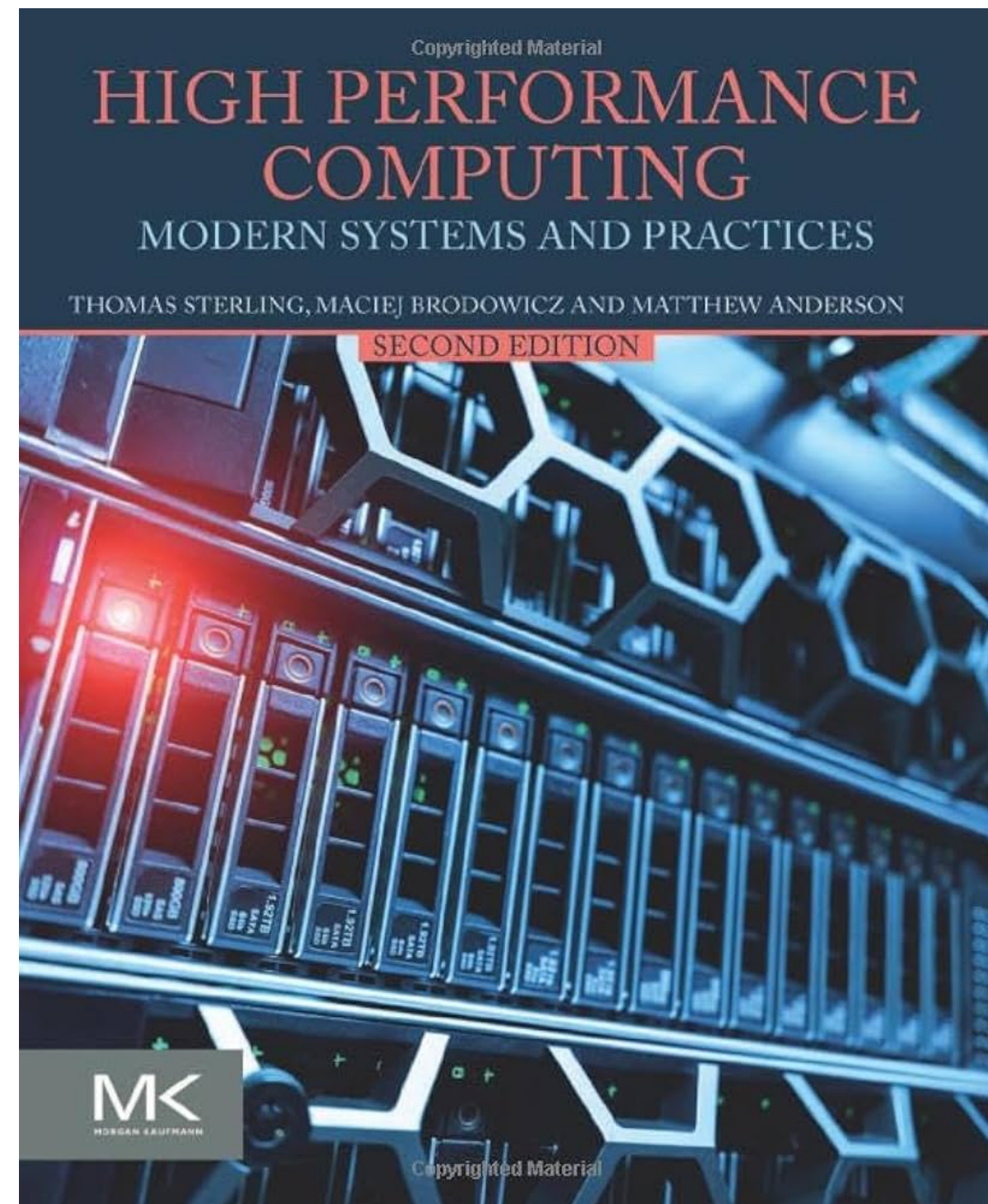
- Projects can explore some of these topics
- Will have some guest lectures on advanced topics

Texts and Resources (don't buy anything)

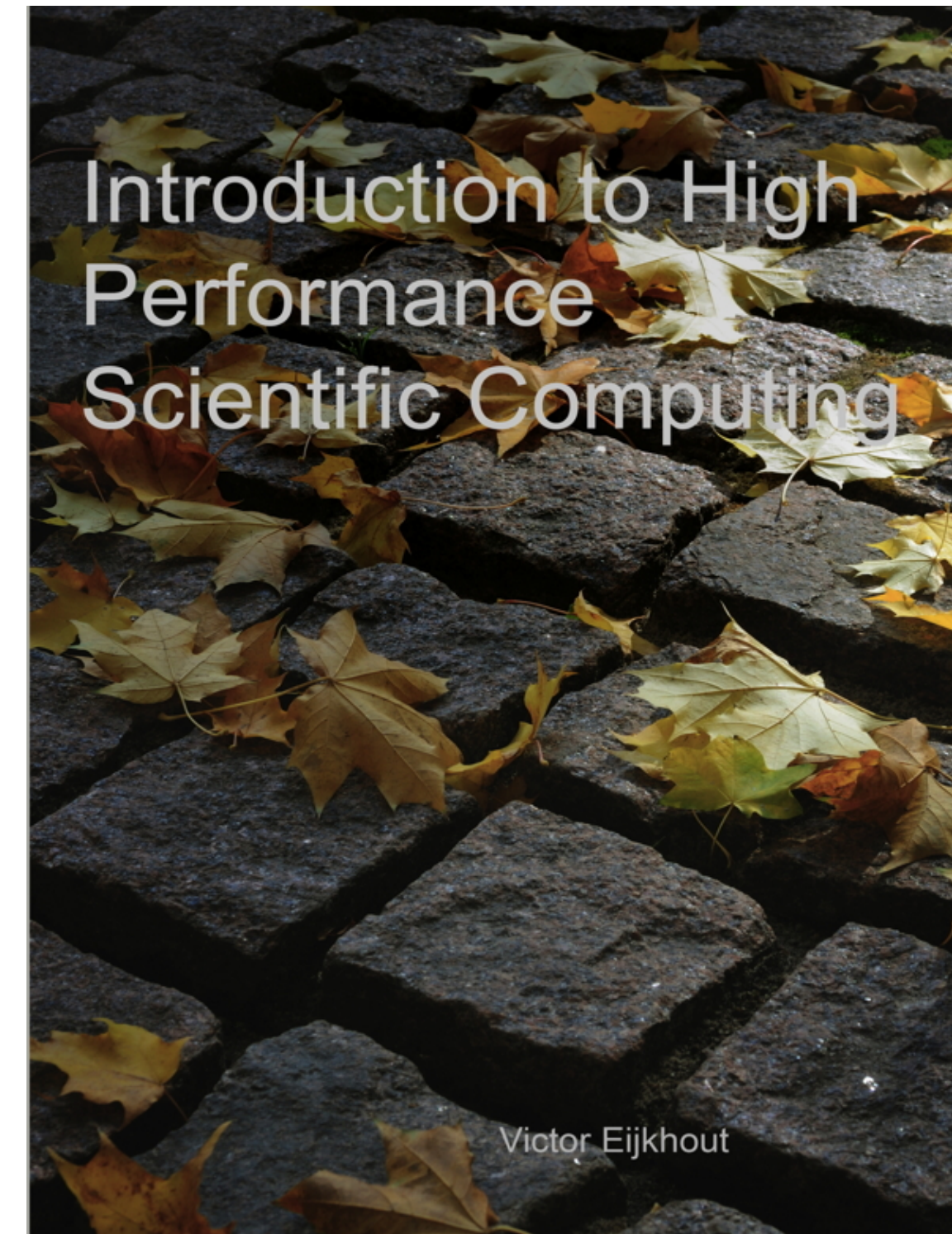


**Primary text for
Numerical PDE's and
Exercises**

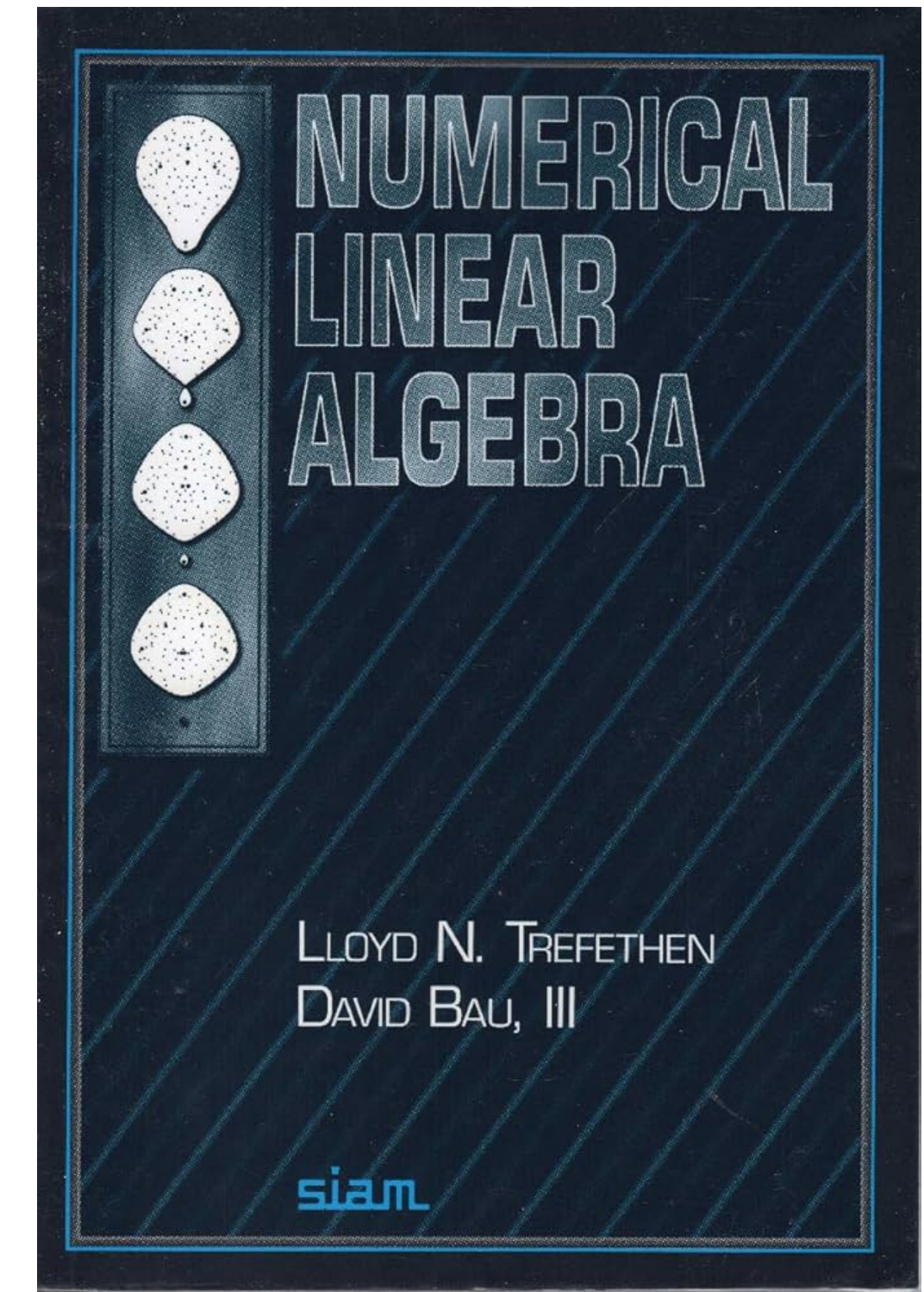
Available as a SIAM
eBook through
Columbia Libraries



Good resource for
general HPC
Ecosystem
Available as an eBook
through Columbia.



4 Volume set on
The Art of HPC
(free pdfs available)



The Bible
Available as a SIAM
eBook through
Columbia Libraries

Texts and Resources (don't buy anything)

Plus innumerable resources on the Web

- PETSc website www.petsc.org
- Stack Overflow
- your favorite chatbot (with care)
- HPC Tutorials
 - [TACC](#) Texas Advanced Computing Center (UT Austin)
 - [Columbia HPC](#)
 - Various HPC centers (National labs)
- GitHub/Gitlab: Lots of open source projects and documentation

Course Logistics

- **Lectures:** 2 per week (will probably move to Mudd 210)
 - some including “tinker time”
 - a few guest lecturers on advanced applications
 - participation expected: 10%
- **Homework** assignments: 40%
 - Mix of analysis and computational work
 - Should have access to Columbia’s Shared Compute Cluster
- **HPC Computational project:** 50%
 - Identify subject early in semester
 - plus possible presentations

Prerequisites

- **Courses:**
 - APMA 4300 — required
 - APMA 4301 — highly recommended
- **Math:**
 - Linear algebra (to 3101, 4007 level) — required
 - Analytic PDE's (to APMA 3102, 4200 level) — highly recommended
- **Computational Skills**
 - Unix/Linux shell/terminal: The command line lives!
 - Compiled Language: C/C++ (preferred), Fortran
 - Python
 - git, ssh
 - make, CMake (useful but not essential)

Questions?

- **Course is capped at 10 students**
- **Make sure this course is right for you:**
- **Fill out User Survey:**
- **Will try to clear the wait list by tonight.**

With serious acknowledgments to Prof David Keyes (KAUST)

- Originated APMA 4302 while Fu Foundation chair Professor of Applied Mathematics in APAM (~2005) (now adjunct Faculty)
- 2x Winner Gordon Bell Prize (1999, 2024)
- Fellow SIAM, AMS, AAAS
- 2024 HPC Wires 35 HPC Legends
- Currently Director of KAUST's Extreme Computing Research Center (ECRC)
- (a fine pianist, and a really nice guy)

